Conley–Zehnder Indices and Bifurcations of the Spatial Rotating Kepler Problem

Dongho Lee

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Seoul National University (SNU), Center for Quantum Structures in Modules and Spaces (QSMS)

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Overview

This talk is based on my thesis and paper [Lee26], which is an extended result of [AFFvK13] dealt with the planar rotating Kepler problem.

1. Basic properties of RKP.
   Motivation, definition, conserved quantities.
2. Periodic orbits of RKP.
   Retrograde, direct, vertical collisions, resonant orbits and their bifurcation structures.
3. Conley–Zehnder indices.
   Computation result, interpretation in Floer theory.
4. (Bonus) Generic Hamiltonian bifurcations.

Rotating Kepler Problem : Basics

Motivation: Circular Restricted Three-body Problem

The **circular restricted three-body problem** (CRTBP) describes a motion of a mass-less object with respect to a gravitational force of two bodies (moon and earth), which are in a motion of planar circular rotation

$$M(t) = (1-\mu)(\cos t, -\sin t, 0), \quad E(t) = -\mu(\cos t, -\sin t, 0).$$

The genuine Hamiltonian is time-dependent, but by imposing a rotating frame, we can write an autonomous Hamiltonian describes CRTBP,

$$H(q,p) = \frac{1}{2}|p|^2 - \frac{\mu}{|q-M|} - \frac{1-\mu}{|q-E|} + (q_1 p_2 - q_2 p_1).$$

where $M = (1-\mu, 0, 0)$, $E = (-\mu, 0, 0)$.

Note. The rotating frame is expressed by adding the angular momentum to the Hamiltonian, $L_3 = q_1 p_2 - q_2 p_1$.

Circular Restricted Three-body Problem

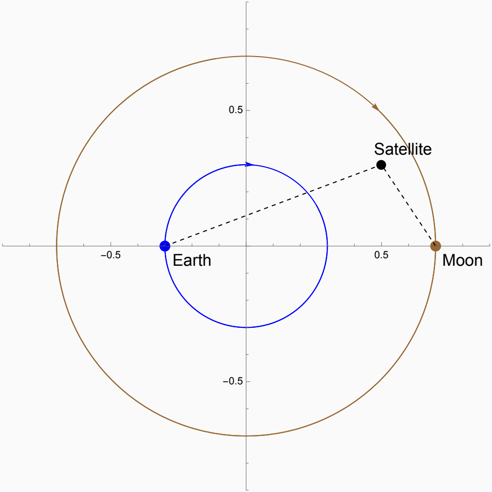


Figure 1: Simple illustration of CRTBP.

Rotating Kepler Problem

Rotating Kepler problem (RKP) is a limit case $\mu = 0$ of CRTBP defined by a Hamiltonian on $T^*\mathbb{R}^3 \setminus \{0\}$,

$$H = E + L_3 = \frac{1}{2}|p|^2 - \frac{1}{|q|} + (q_1 p_2 - q_2 p_1).$$

E : **Kepler energy**, describes Kepler problem (two-body system).

H : **Jacobi energy**. (usually, we write the value of H by c.)

This describes a motion of a mass-less object under a gravitational force from the origin in a rotating frame. (e.g. a satellite near the Earth.)

Why is RKP important?

- Solution of CRTBP is impossible to compute analytically.
- Since RKP is *completely integrable*, we have complete solutions.
- Regarding CRTBP as a perturbed system of RKP with parameter μ, we can find out some features of CRTBP for small mass ratio.
- For example, periodic orbits of RKP works as a backbone of CRTBP: the *retrograde*, *direct* and *vertical collision* orbits. (cf. [Poi87].)

Hill's Region

The Hamiltonian of RKP can be written as

$$H = \frac{1}{2}|\tilde{p}|^2 - \left(\frac{q_1^2 + q_2^2}{2} + \frac{1}{|q|} \right) = \frac{1}{2}|\tilde{p}|^2 + U(q),$$

where $\tilde{p} = (p_1 - q_2, p_2 + q_1, p_3)$. We call $U(q)$ an **effective potential**.

Notice that if $H(q,p) = c$, then $U(q) \leq c$.

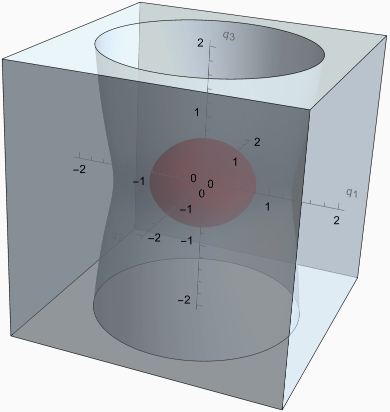
Hill's region : $\mathfrak{R}_c = \left\{ q \in \mathbb{R}^3 \, : \, U(q) \leq c \right\} = \left\{ q \in \mathbb{R}^3 \, : \, H(q,p) = c \right\}.$

The orbit with Jacobi energy c always remains in $\mathfrak{R}_c$.

- $\mathfrak{R}_c$ has 2 components (1 bounded, 1 unbounded) if $c < -3/2$.
- $\mathfrak{R}_c$ has 1 unbounded component if $c \geq -3/2$.

We call $c = -3/2$ a **critical energy**, and focus on the bounded component of $\mathfrak{R}_c$ for $c < -3/2$.

Hill's Region



The Hill's region of RKP with energy $c < -3/2$, which has 2 components.

Kepler's Law

Let $E = |p|^2/2 - 1/|q| < 0$ be the Kepler Hamiltonian. We call the Hamiltonian orbits of E a **Kepler orbit**.

Kepler established these 3 laws in 17C, *based on pure observations!*

1. Kepler orbits are ellipses with one focus at the origin.
2. The angular momentum $L = q \times p$ is a conserved quantity.
3. The period τ is determined by the Kepler energy,

$$\tau = \frac{2\pi}{(-2E)^{3/2}}.$$

Conserved Quantities

Consider the Hamiltonian equation

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}.$$

We interpret this as a **Hamiltonian vector field**

$$X_H = \sum_i \frac{\partial H}{\partial p_i}\frac{\partial}{\partial q_i} - \frac{\partial H}{\partial q_i}\frac{\partial}{\partial p_i}.$$

A function F is a **conserved quantity** if $X_H(F) = 0$, i.e.,

$$\{F, H\} := \frac{\partial F}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial F}{\partial p_i}\frac{\partial H}{\partial q_i} = \frac{\partial F}{\partial q_i}\dot{q}_i + \frac{\partial F}{\partial p_i}\dot{p}_i = 0.$$

We call $\{F, H\}$ a **Poisson bracket**, which measures the conservation.

Conserved Quantities

Kepler problem has six well-known conserved quantities (two 3-vectors):

Angular momentum $L = q \times p$.

- Direction of L = Normal to the plane which the orbit contained in.

Laplace–Runge–Lenz vector $A = p \times L - \frac{q}{|q|}$

- Direction of A = Direction of the major axis of the ellipse.

We have $\{E, L_i\} = \{E, A_i\} = \{L_i, A_i\} = 0$ for $i = 1, 2, 3$.

Another relation: $2E|L|^2 + 1 = |A|^2$.

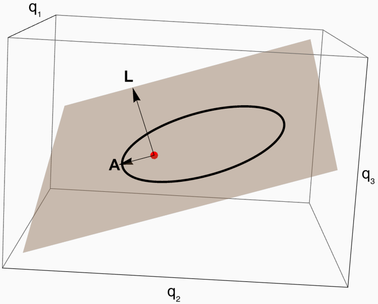
Note 1. The pair (E, L_i, A_i) provides an *action-angle coordinates*, which allows us to write down the explicit solutions in a linear motion.
(cf. Arnold–Liouville theorem, [Arn89].)

Note 2. For RKP, $H = E + L_3$, so (E, L_3, A_3) are conserved quantities.

Conserved Quantities

A Kepler orbit is determined by E, L and A, whose formula in the polar coordinate of $L \cdot q = 0$ is

$$r = \frac{|L|^2}{1 + |A| \cos(\theta - g)} \quad (g \text{ is determined by the direction of } A).$$



The eccentricity is given by $\varepsilon^2 = |A|^2 = 2E|L|^2 + 1$.

Periodic Orbits of RKP

Since $\{E, L_3\} = 0$, we can decompose the Hamiltonian flow of RKP by

$$Fl_t^H = Fl_t^E \circ Fl_t^{L_3}.$$

Which means that an orbit of RKP is a composition of two motions,

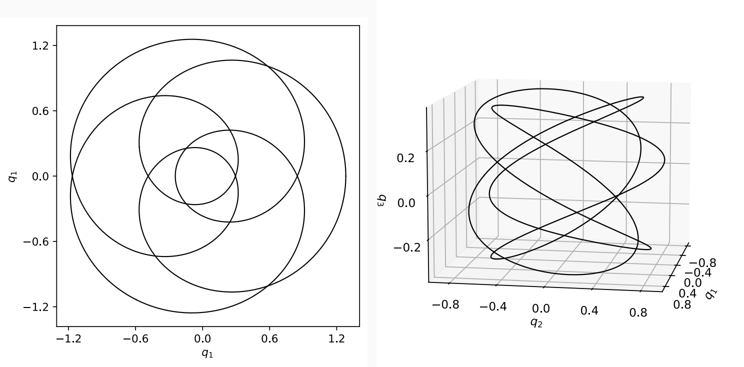
- Fl_t^E: Elliptic rotation in a space with period $2\pi/(-2E)^{3/2}$.
- $Fl_t^{L_3}$: Rotation along q_3-axis with period 2π.

For general elliptic orbits, a resonance condition of two periods is required to be periodic,

$$k\frac{2\pi}{(-2E)^{3/2}} = 2\pi l \quad \Rightarrow \quad E = E_{k,l} := -\frac{1}{2}\left(\frac{k}{l}\right)^{2/3}.$$

Thus if $E = E_{k,l}$, every Kepler orbit is periodic in RKP. We call this a **resonant orbit** of type $\Sigma_{k,l}$.

Resonant Solutions



These are how planar and spatial resonant orbits look like.
(Figure in courtesy of Chankyu Joung.)

Note. If we consider a fixed Jacobi energy c, then the angular momentum of $\Sigma_{k,l}$-orbit is also characterized by $L_3 = c - E_{k,l}$.

Retrograde and Direct Orbits

If a Kepler orbit is *planar* and *circular*, the composition with L_3-flow cannot effect the periodicity of the orbit.

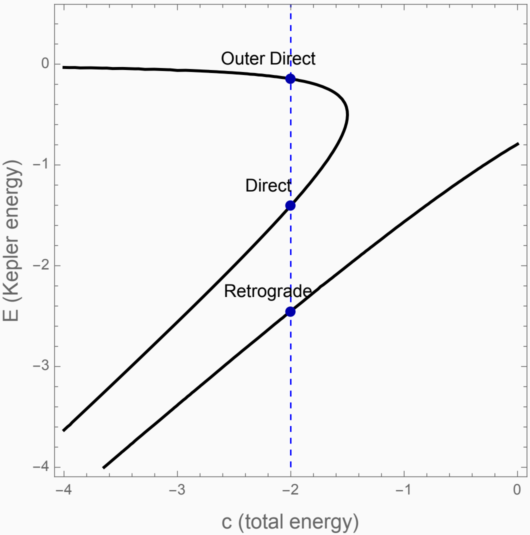
- Planar condition : $L = L_3$.
- Circular condition: $\varepsilon^2 = 2EL_3^2 + 1 = 2E(c - E)^2 + 1 = 0$.

For fixed $c < -3/2$, there are 3 **planar circular orbits** with different E.

1. **Retrograde orbit** γ_+: $L_3 > 0$, smaller E and smaller radius.
2. **Direct orbit** γ_-: $L_3 < 0$, larger E and larger radius.
3. The rest one, outer direct orbit, lies on the unbounded component, and not of our interest.

Note. $\gamma_\pm$ can be continued to CRTBP.

Retrograde and Direct Orbits



Collision Orbits

[Mos70] For $H < -3/2$, we can embed the Hamiltonian flow on the level set $H^{-1}(c)$ into a flow on T^*S^3 by **Moser regularization**.

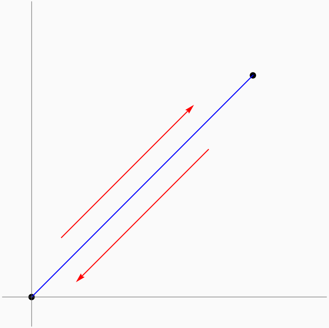
For Kepler problem, it is embedded as a unit geodesic flow on S^3.

[CFvK14] showed that RKP is embedded as a Finsler geodesic flow.

$\Rightarrow$ Compactification of the energy level set by $ST^*S^3 \simeq S^3 \times S^2$.

Collision orbits

Collision orbits are added. $(\varepsilon = |A| = 1,\ L = 0,\ E = c.)$



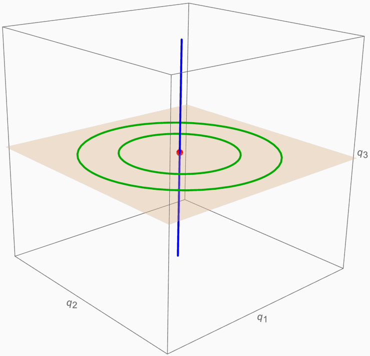
This orbit bounds between two points; a shrunken ellipse.

Period is also determined by $\tau = 2\pi/(-2E)^{3/2}$.

$\Rightarrow$ General collision orbits are not periodic in RKP except $\Sigma_{k,l}$-types.

Vertical Collision Orbits

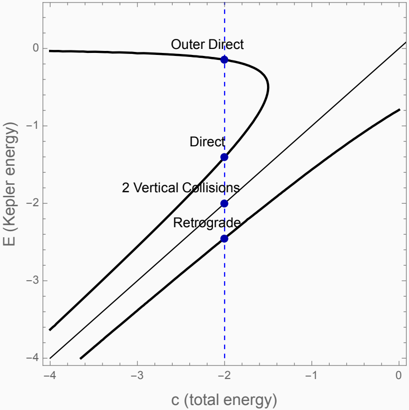
Vertical collision orbits $\gamma_{c_\pm}$ are not effected by L_3-flow, so it remains periodic in RKP.



Note. $\gamma_{c_\pm}$ are continued to *halo orbits* of CRTBP.

Vertical Collision Orbits

Since $L_3 = 0$ $E = c$, we can mark the vertical collision orbits as follows.



Space of Periodic Orbits of RKP

Moduli Space of Kepler Orbits

Recall. E, L and A characterizes the Kepler orbit.

Denote $x = \sqrt{-2E}L - A$, $y = \sqrt{-2E}L + A$.

$\Rightarrow |x|^2 = |y|^2 = -2E|L|^2 + |A|^2 = 1$.

The moduli space of the Kepler orbits with Kepler energy E is

$$\mathcal{M}_E = \left\{(x, y) \,:\, |x|^2 = |y|^2 = 1\right\} \simeq S^2 \times S^2.$$

In Moser regularization viewpoint, we have

(Space of unit geodesics of S^3) $= ST^*S^3/S^1 \simeq S^2 \times S^2$.

Note. In the planar problem, the moduli space is $ST^*S^2/S^1 \simeq S^2$.

Properties of $\mathcal{M}_E$

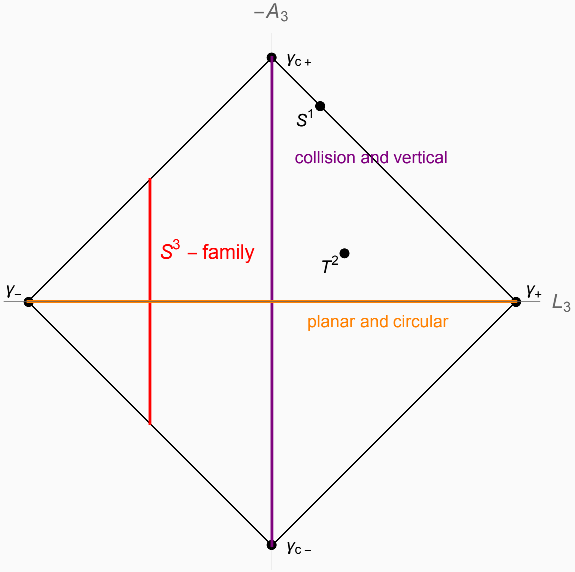
$L_3 = (x_3 + y_3)/\sqrt{-2E}$ and $A_3 = (x_3 - y_3)/\sqrt{-2E}$ can be regarded as a function on $\mathcal{M}_E$.

1. $\gamma_\pm = ((0,0,\pm 1),(0,0,\pm 1))$ are minimum and maximum of L_3.
2. $\gamma_{c_\pm} = ((0,0,\pm 1),(0,0,\mp 1))$ are minimum and maximum of A_3.
3. Every regular level set of L_3 and A_3 is S^3. (Handle attachment)
4. $L_3^{-1}(0)$ is singular, homeomorphic to a suspension of S^2.

$\Rightarrow$ Generally, $\Sigma_{k,l} = L_3^{-1}(c - E_{k,l})$ is topologically S^3.

Note. Only finite number of orbits in $\Sigma_{k,l}$ survives in CRTBP.

A Diagram of $\mathcal{M}_E$



Bifurcation Structure

Let $c_{\pm}^{k,l} = E_{k,l} \pm (-2E_{k,l})^{-1/2}$, the Jacobi energy of retrograde/direct orbits with Kepler energy $E_{k,l}$.

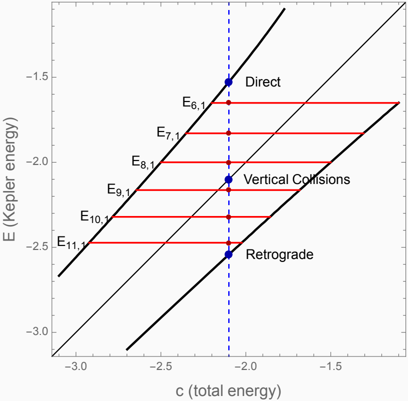
Here's an evolution of $\Sigma_{k,l}$-family with increasing c.

1. At $c = c_{-}^{k,l}$, $\Sigma_{k,l}$ degenerates to 1 orbit; a multiple cover of γ_{-}.
2. Consider the resonance condition: $k\tau = 2\pi l$. The Kepler flow rotates k-times, while L_3-flow rotates l-times. In total, we have $(k-l)$-rotation, so we have $\gamma_{-}^{k-l} = \Sigma_{k,l}$.
3. For $c_{-}^{k,l} < c < c_{+}^{k,l}$ (except $c = E_{k,l}$), we have S^3-family.
4. Similarly, at $c = c_{+}^{k,l}$ we have $\gamma_{+}^{k+l} = \Sigma_{k,l}$.

In short, as c varies, $\Sigma_{k,l}$ is born at γ_{-}^{k-l} and dies at γ_{+}^{k+l}.

Periodic Orbits in $H^{-1}(c)$

For generic energy level c, the energy hypersurface $H^{-1}(c)$ contains 4 nondegenerate orbits and infinitely many S^3-families.



To see bifurcation, we can see this graph in horizontal direction.
Each red line is identical to $\mathcal{M}_{E_{k,l}}$.

Conley–Zehnder Indices

Conley–Zehnder Index and Bifurcation

The **Conley–Zehnder index** is, heuristically, a symplectic invariant of a periodic orbit which measures a winding number of a symplectic frame, which is given by an integer $\mu_{CZ}(\gamma) \in \mathbb{Z}$.

Let Ψ_t be the linearization Hamiltonian flow, which is a path of symplectic matrices.

$\mu_{CZ}(\gamma)$ is well-defined if $\det(\Psi_\tau - \mathrm{Id}) \neq 0$ where τ is the period of γ. This condition is called **non-degeneracy** of γ.

Bifurcation of Periodic Orbits

Let γ be a periodic orbit with energy c, and consider a change of the energy level near c.

1. If γ is non-degenerate at c_0, then there exists an *orbit cylinder* Γ parametrized by $(c_0 - \varepsilon, c_0 + \varepsilon)$. That is an embedded cylinder

$$\begin{aligned} \Gamma : S^1 \times (c_0 - \varepsilon, c_0 + \varepsilon) &\to W \\ (t, c) &\mapsto \gamma_c(t) \end{aligned}$$

   where γ_c is a periodic orbit with energy c.
2. CZ index is constant along Γ_c.
3. When γ_c become degenerate, the orbit cylinder breaks: the cylinder might vanish, or the other periodic orbit might emergy.
4. This is called **bifurcation**: birth or death of periodic orbits.
5. CZ index changes exactly when γ_c degenerates.

What is Floer homology?

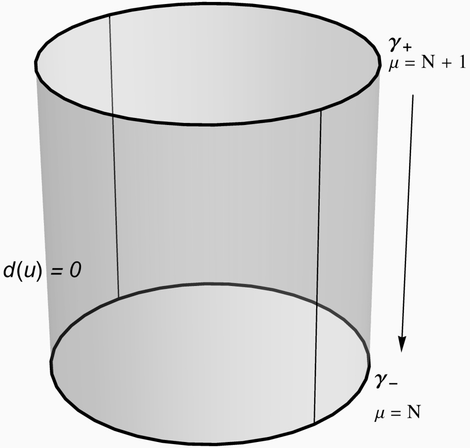
A **Hamiltonian Floer homology** is a homology defined on a symplectic manifold W by a chain complex $CF(W, H)$, where

- generators are 1-periodic orbits, graded by Conley–Zehnder indices.
- differential is defined by counting of *Floer cylinders*, which are solutions of a PDE with boundaries asymptotic to periodic orbits.

Fact. In nice cases, if W is compact without boundary, $HF_*(W, H)$ is isomorphic to $H_*(W)$. (independent of H!)

This can be regarded as a Morse homology on a free loop space with *symplectic action functional*. See [AD14] for details.

What is Floer homology?



Differential of a Floer chain complex.

Variants of Floer Homology

1. **Local Floer homology**

We can compute $HF(H)$ on a small neighborhood of a specific periodic orbit γ, which does not contain any other orbits.

This is invariant under small perturbation of the Hamiltonian (cf.[GG10]).

$\Rightarrow$ A symplectic invariant which can track the bifurcation.

2. **Symplectic homology**

For open symplectic manifold, we define a **symplectic homology** $SH(W)$ as a direct limit of Hamiltonian Floer homology.

Periodic orbits in a specific energy hypersurface $H^{-1}(c)$ can be treated in this way, where $SH(W)$ consists of periodic orbits of *any* period.

Fact. [Vit99] $SH_*(T^*M;\mathbb{Q}) \simeq H_*(\mathcal{L}M, \mathbb{Q})$ with a degree shift.

Conley–Zehnder Index of Planar Circular Orbits

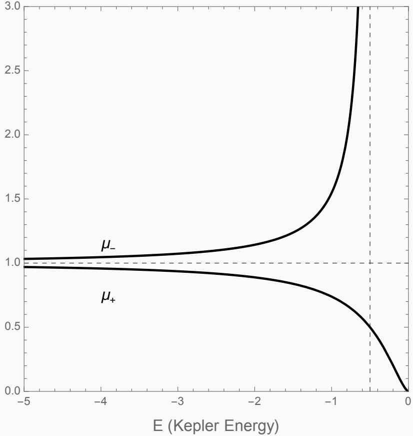
Theorem

Let $\gamma_\pm$ be the retrograde and direct orbits of Kepler energy E where $E \neq E_{k,l}$ for any k, l. Then $\gamma_\pm$ and their multiple covers are non-degenerate. The Conley–Zehnder index of N-th iterate of $\gamma_\pm$ is

$$\begin{aligned}\mu_{CZ}(\gamma_\pm^N) &= 2 + 4\max\left\{ n \in \mathbb{Z}_{>0} : n < N\frac{(-2E)^{3/2}}{(-2E)^{3/2} \pm 1} \right\} \\ &= 2 + 4\left\lfloor N\frac{(-2E)^{3/2}}{(-2E)^{3/2} \pm 1} \right\rfloor := 2 + 4\lfloor N\mu_\pm(E) \rfloor.\end{aligned}$$

The index is exactly the twice compare to the planar problem, which was computed in [AFFvK13].

Conley-Zehnder Index of Planar Circular Orbits



The index of $\gamma_\pm^N$ is initially $4N \mp 2$, and changes by ∓ 4 whenever $\mu_\pm$ touches $k/N \Leftrightarrow E = E_{N\mp k,k}$. (Bifurcation occurs)

Conley-Zehnder Index of Vertical Collision Orbits

Theorem

Let $\gamma_{c_\pm}$ be the vertical collision orbits of Kepler energy E where $E \neq E_{k,l}$ for any k, l. Then $\gamma_{c_\pm}$ and their multiple covers are non-degenerate. The Conley-Zehnder index of N-th iteration of $\gamma_{c_\pm}$ is

$$\mu_{CZ}(\gamma_{c_\pm}^N) = 4N.$$

We don't have $\mu_{CZ}(\gamma^N) = N\mu_{CZ}(\gamma)$ in general. This is due to highly symmetric structure of vertical collision orbits.

Summarization of the result

1. Retrograde orbit γ_+^N: $\mu = 4N - 2$ for low energy, and decreases by 4 whenever $c = c_{N-k,k}^+$, eventually becomes 2.
   In particular, a simple retrograde orbit has always index 2, and does not bifurcate at all.
2. Direct orbit γ_-^N: $\mu = 4N + 2$ for low energy, and increases by 4 whenever $c = c_{N+k,k}^-$, diverges as $c \to -3/2$.
3. Vertical collisions $\gamma_{c_\pm}^N$: $\mu = 4N$ for any energy.

Interpretation by Symplectic Homology

$$SH_*^{+,S^1}(T^*S^3;\mathbb{Q}) \simeq \begin{cases} \mathbb{Q} & * = 2, \\ \mathbb{Q}^2 & * = 2k \geq 4, \\ 0 & \text{otherwise.} \end{cases}$$

For fixed N, there exists $c \ll -3/2$ such that $H^{-1}(c)$ consists of

1. $k(\leq N)$-th covers of $\gamma_\pm$ of index $4k \mp 2$. (No bifucation)
2. Higher covers have index $> 4N + 2$.

Up to degree $4N + 2$, we have

1. One generator at degree 2. (γ_+.)
2. Two generators at degree 6, 10, 14, $\cdots$, $4N + 2$. (γ_+^{k+1} and γ_-^k.)
3. Two generators at degree 4, 8, 12, $\cdots$, $4N$. ($\gamma_{c_+}^k$ and $\gamma_{c_-}^k$.)

This describes $SH_*^{+,S^1}(T^*S^3)$ up to certain degree completely.

Morse-Bott Property

$\Sigma_{k,l}$-families: We use **Morse-Bott spectral sequence**.

$\Rightarrow$ We need Morse-Bott property (kind of non-degeneracy), and must compute the linearized return map.

We used two different action-angle coordinates:

1. Delaunay coordinate : $(p_l, p_g, p_\theta) = (1/\sqrt{-2E}, |L|, L_3)$.

   Works for planar problem ([AFFvK13]), but degenerates at every planar orbit in the spatial case.

2. **LRL coordinate** : $(p_l, p_\eta, p_\theta) = (1/\sqrt{-2E}, A_3, L_3)$.

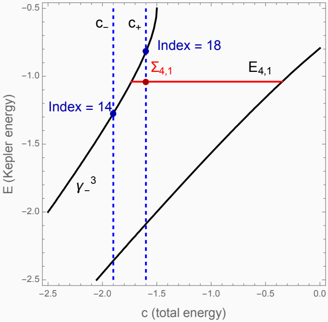
   Also degenerates at some orbits, but covers planar orbits.

Index of Degenerate Orbits

Theorem

Index of S^3-family $\Sigma_{k,l}$ with Kepler energy $E_{k,l}$ is

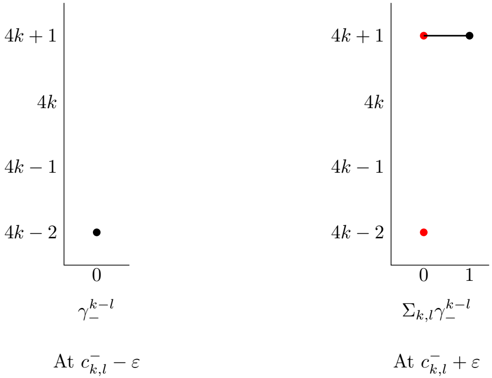
$$\begin{aligned}\mu_{RS}(\Sigma_{k,l}) &= \mathrm{shift}(\Sigma) + \dim S^3/2 \\ &= (4k-2) + 3/2 = 4k - 1/2.\end{aligned}$$



Index of Degenerate Orbits

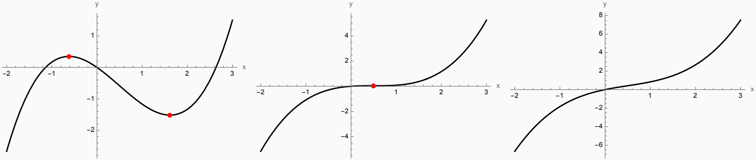
Previous results + local invariance of the symplectic homology.

We use *Morse–Bott spectral sequence* for the argument.

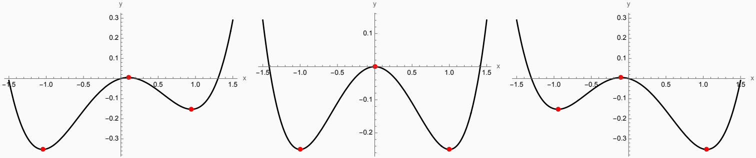


Bonus : Generic Hamiltonian Bifurcations

Generic Bifurcations of Morse Function



Birth-death bifurcation of critical points ($f_t(x) = x^3 - x^2 + tx$).



Handle-slide of critical points ($f_t(x) = x^4 - x^3 - x^2 + tx$).

Generic Bifurcations of Periodic Orbits

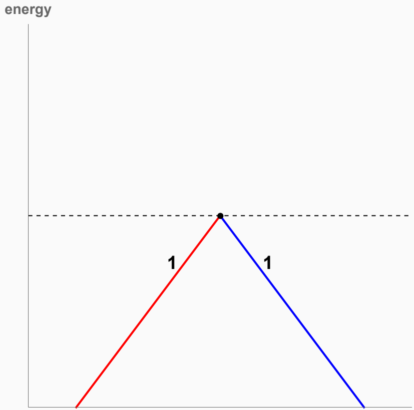
Meyer [Mey70] classified generic bifurcations of Hamiltonian periodic orbits into 5 types. (cf. [AM78] Ch.8)

1. Bifurcation occurs when the orbit cylinder breaks, i.e. there exists a eigenvalue of linearized return map such that $\lambda = 1$.
2. If $\lambda^k = 1$, k-th cover of the orbit bifurcates.
3. Find fixed points of an iterated return map. (Birkhoff normal form)

Note. The case of Hamiltonian orbit is different from the one of Morse function, because of multiple covers.

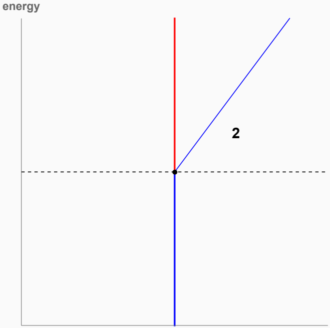
Birth-death

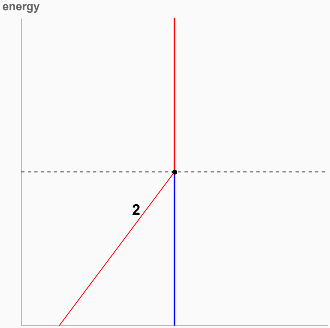
Birth-death: bifurcation of simple orbits.



Period Doubling and Materialization

Period doubling / Materialization: bifurcation of double cover.

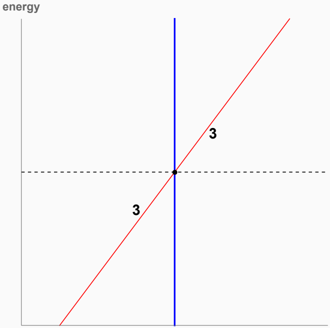


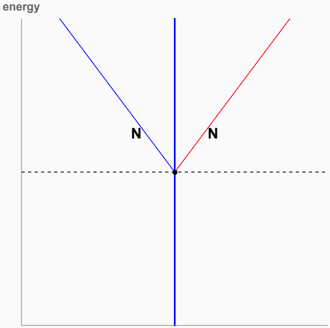


Phantom Kiss and Emission

Phantom kiss (or touch-and-go): bifurcation of triple or 4-th cover.

Emission : bifurcation of $N(\geq 4)$-th cover.





If the system possesses a symmetry, there are other types.

Ex. CRTBP has $\mathbb{Z}_2$-symmetry, which leads to a pitchfork bifurcation.

Local Floer Chain Complex

Recall. Degeneracy of an orbit = bifurcation = change of μ_{CZ}.

However, $HF_{loc}(W)$ is invariant through a bifurcation.

$\Rightarrow$ Bifurcation occurs a mutation of a chain complex.

- The number of generators changes.
- The degree (= CZ index) also changes.
- Differential map is re-arranged to make the homology consistent.

The simplest example: Two orbits of the birth-death must have index difference 1 to cancel each other.

Goal: Describe the explicit change of the chain complex through generic bifurcation types. (ongoing work with Hong-kwon Cho (SNU)).

Thank you for your attention!

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